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Parity-Selective Quantum Interference in Unconventional Magnets

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Nonrelativistic spin splitting has emerged as a defining feature of unconventional magnetic states. Here we show that quantum interference offers a symmetry-resolved probe for the parity of such spin splitting, as even- and odd-parity magnets exhibit qualitatively different weak (anti-)localization behaviors. Even-parity magnets can host time-reversal symmetric scattering loops depicted by the spin group, giving rise to a weak localization that can be suppressed by spin-orbit coupling. In contrast, odd-parity magnets have a hidden weak antilocalization that can be released by spin-orbit coupling. We further derive the corresponding magnetoconductivity formulas to facilitate experimental explorations. Our results not only advance the weak (anti-)localization theory to unconventional magnets but also suggest a practical diagnostic of unconventional spin-split magnetic states.

Introduction - Non-relativistic spin splitting [1–4] magnet depicted by spin space groups [5–7] has emerged as a paradigm in magnetism. Recent studies have identified a diverse library of candidate materials, classified primarily according to the parity of spin splitting in momentum space: even-parity altermagnets (e.g., d-wave, g-wave, ...) [8–21] and odd-parity magnets (e.g., p-wave, f-wave, ...) [21–30]. Nevertheless, compelling experimental signature of these unconventional magnets is still a topic of tremendous interest [31–40]. Symmetry can always give decisive criteria for states of condensed matter. Weak localization (WL) and weak antilocalization (WAL) are highly sensitive to symmetry, appearing as low-temperature positive and negative magnetoconductivity, respectively, in metals [41], semiconductors [42–45], 2D materials [46–49], and topological materials [50–60]. However, weak (anti-)localization arise from the quantum interference between time-reversal scattering loops, which are highly nontrivial to be defined in magnets [61, 62].

In this Letter, we investigate the weak (anti-)localization in even- and odd-parity magnets, focusing on the definition of time-reversal symmetry in the context of spin groups and influence of spin-orbit coupling. In the absence of spin-orbit coupling, we figure out the spin-group symmetries that support the time-reversal scattering loops and WL and WAL channels in even- and odd-parity magnets. Specifically, even-parity d-wave altermagnets can host two WL channels that add up to each other, while odd-parity p-wave magnets have a WL channel and a WAL channel that exactly cancel each other. Hence, we predict WL in altermagnets and no WL or WAL in p-wave magnets in the absence of spin-orbit coupling. Spin-orbit coupling can suppress the WL channels in both d-wave altermagnets and p-wave magnets, leading to the suppression of WL in d-wave altermagnets and activation of WAL in p-wave magnets in the presence of spin-orbit coupling, as summarized in Figure 1 and Table I. More importantly, our conclusions also apply to higher-order even- and odd-parity magnets (e.g., f-, g- or i-wave magnets). Our results advance the weak (anti-)localization theory and will be helpful for explorations on unconventional magnets.

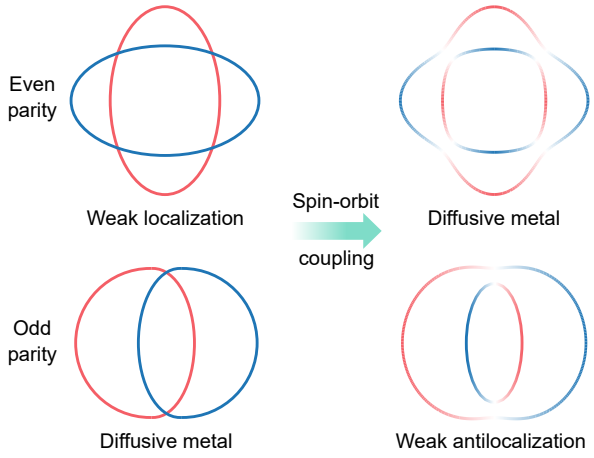


FIG. 1. The spin-up (red) and spin-down (blue) Fermi surfaces of even-parity altermagnets and odd-parity p-waves magnets in the absence and presence of spin-orbit coupling. In absence of spin-orbit coupling, altermagnets may exhibit weak localization (WL) and p-wave magnets are diffusive metals. Spin-orbit coupling can suppress WL in altermagnets and release weak antilocalization (WAL) in p-wave magnets.

TABLE I. Channels of weak localization (WL) and weak antilocalization (WAL) of even- and odd-parity magnets and their spin-group symmetries in the absence and presence of spin-orbit coupling (SOC). We adopt the notation of spin group $\mathcal{T}[A||B]$, $\mathcal{T}$ is the overall time-reversal operation, A is for spin degree of freedom, B is for real space, $C_{2x(y)}$ is two-fold spin rotation around the x (y) axis, and E is identity. The + and − signs mean that the channels add up or cancel each other.

		without SOC	with SOC
Even	$\mathcal{T}[C_{2x}\|\|E]$ (WL)	✓	✗
	+		
	$\mathcal{T}[C_{2y}\|\|E]$ (WL)	✓	✗
Odd	$\mathcal{T}[C_{2z}\|\|E]$ (WL)	✓	✗
	−		
	$\mathcal{T}$ (WAL)	✓	✓

Magnets with even- and odd-parity spin splitting - The Hamiltonian that describes electrons in 2D magnets with even- and odd-parity non-relativistic spin splitting can be given as

$$H(\mathbf{k}) = ak^2 + m_{\mathbf{k}}\sigma_z + \lambda(k_x\sigma_y - k_y\sigma_x), \tag{1}$$

where the a term represents the spin-independent parabolic band and $k^2 = k_x^2 + k_y^2$, λ measures the spin-orbit coupling, and $m_{\mathbf{k}}$ describes the non-relativisitic spin spitting with different orders of symmetries (d, g, i for even-parity and p, f for odd-parity) [63]

$$\begin{aligned} m_{\mathbf{k}}^p &= Jk_x, \\ m_{\mathbf{k}}^d &= J(k_x^2 - k_y^2), \\ m_{\mathbf{k}}^f &= Jk_x(k_x^2 - 3k_y^2), \\ m_{\mathbf{k}}^g &= Jk_xk_y(k_x^2 - k_y^2), \\ m_{\mathbf{k}}^i &= Jk_xk_y(k_x^2 - 3k_y^2)(k_y^2 - 3k_x^2). \end{aligned} \tag{2}$$

For instance, altermagnets are of even parity and p-wave magnets are of odd parity. Because unconventional magnets with non-relativistic spin splitting are antiferromagnets with zero net magnetization, the integral of $m_{\mathbf{k}}$ on any isoenergy contour is supposed to be zero.

Time-reversal symmetry in context of spin group - WL or WAL arises from quantum interference between time-reversed loops formed by trajectories of electrons as they are scattered by disorder in solids [64–66], so time-reversal symmetry is required. However, it is always nontrivial to define time-reversal symmetry in magnets. We will show that, with the help of spin group, we can introduce the anti-unitary operators to depict the effective time-reversal symmetries for the unconventional magnets with non-relativistic spin splitting.

For $m_{\mathbf{k}}$ of even parity (e.g., altermagnets), if in the absence of spin-orbit coupling ($\lambda = 0$), there are two anti-unitary operators

$$\mathcal{T}_1 = \sigma_z K, \quad \mathcal{T}_2 = K, \tag{3}$$

with K the complex conjugate, so that $\mathcal{T}_1 H(\mathbf{k})\mathcal{T}_1^{-1} = H(-\mathbf{k})$ and $\mathcal{T}_2 H(\mathbf{k})\mathcal{T}_2^{-1} = H(-\mathbf{k})$. They depict the effective time-reversal symmetries

$$\mathcal{T}[C_{2x}||E], \quad \mathcal{T}[C_{2y}||E], \tag{4}$$

respectively, of altermagnets [8] in the absence of spin-orbit coupling, where $\mathcal{T}$ is the overall time-reversal operation and $[C_{2x/2y}||E]$ is the spin group operation that rotates spin by π around the x/y axis and remains identity in real space. Because $\mathcal{T}_1^2 = 1$ and $\mathcal{T}_2^2 = 1$, they belong to the orthogonal class according to the symmetry classification of disordered ensembles [67]. The quantum interference of the orthogonal class is WL [68]. Spin-orbit coupling forbids the free spin rotation so it breaks $\mathcal{T}_1$ and $\mathcal{T}_2$, leading to the suppression of WL.

For $m_{\mathbf{k}}$ of odd parity (e.g., p-wave magnets), if in the absence of spin-orbit coupling ($\lambda = 0$), there are two anti-unitary operators

$$\mathcal{T}_3 = \sigma_x K, \quad \mathcal{T}_4 = \sigma_y K, \tag{5}$$

so that $\mathcal{T}_3 H(\mathbf{k})\mathcal{T}_3^{-1} = H(-\mathbf{k})$ and $\mathcal{T}_4 H(\mathbf{k})\mathcal{T}_4^{-1} = H(-\mathbf{k})$. They depict the effective time-reversal symmetries

$$\mathcal{T}[C_{2z}||E], \quad \mathcal{T} \tag{6}$$

respectively, in the absence of spin-orbit coupling. Because $\mathcal{T}_3^2 = 1$, it also belongs to the orthogonal class that gives rise to WL. In contrast, $\mathcal{T}_4^2 = -1$ and it belongs to the symplectic class [67]. The quantum interference of the symplectic class is WAL [68]. The weak localization from $\mathcal{T}_3$ cancels with weak antilocalization from $\mathcal{T}_4$, leading to vanishing WL or WAL in the absence of spin-orbit coupling [69]. Spin-orbit coupling forbids the free spin rotation so it breaks $\mathcal{T}[C_{2z}||E]$ but preserves $\mathcal{T}$, therefore, WL is suppressed while WAL is released by spin-orbit coupling.

Feynman diagrams and Cooperon gaps - Now we use diagrammatic calculations [64, 65] to verify the above physical picture. Although the calculation is routine, the application to the magnets with non-relativistic spin splitting is nontrivial. Different from the Drude conductivity arising from the particle-hole correlation, the quantum interference is from particle-particle correlation, similar to the Cooper pairing of electrons, so the quantum interference is also referred to as the diffusion of Cooperons. The electron has a spin of 1/2, so as two paired electrons, a Cooperon can have the total angular momenta of 1 and 0, i.e., the triplet and singlet Cooperons [70]. Each of the triplet Cooperons gives a WL channel and the singlet Cooperon gives one WAL channel. Because the triplet Cooperons have a finite angular momentum of 1, they

can be suppressed by the spin-orbit coupling which behaves like a spin-dependent "magnetic field". The singlet Cooperon therefore is robust against spin-orbit coupling because it has no angular momentum.

In the language of diagrammatic techniques, the suppression of a WL or WAL channel is described by a finite Cooperon gap. A vanishing Cooperon gap leads to a divergent contribution to the conductivity to support a WL or WAL channel. The Cooperons are calculated with the help of the diagrams of Bethe-Salpeter equation in Fig. 2(a), where the arrow solid lines are the disorder-averaged retarded/advanced Green's functions defined in the σ_z-eigenbasis [71–73] as $G^{R/A}_{\mathbf{k}} = (g^{R/A}_{\mathbf{k}})^{-1}[E_F - ak^2 + m_{\mathbf{k}}\sigma_z + \lambda(k_x\sigma_y - k_y\sigma_x)]$ and $g^{R/A}_{\mathbf{k}} = (E_F \pm i\Delta - ak^2)^2 - m^2_{\mathbf{k}} - \lambda^2k^2$, E_F is the Fermi energy, $\Delta = \Gamma_0\rho\pi/2$ is the disorder averaged self-energy with Γ_0 the bare Cooperon and ρ the density of states at the Fermi energy. The Bethe-Salpeter equation of the full Cooperon $\Gamma(\mathbf{q})$ reads

$$\Gamma(\mathbf{q}) = \left[\Gamma_0^{-1} - \Pi(\mathbf{q})\right]^{-1}, \tag{7}$$

where the Kernel function $\Pi(\mathbf{q}) = \sum_{\mathbf{k}} G^R_{\mathbf{k}} \otimes G^A_{\mathbf{q}-\mathbf{k}}$. The gaps of Cooperon in different channels can be evaluated from $[\Gamma(\mathbf{q})]^{-1}$ as $\mathbf{q} \to 0$. Above we have used the particle-particle basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$. More conveniently, they can be transformed into the singlet and triplet basis $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle, |0,0\rangle\}$ via a transformation $\mathcal{U}$ [69].

For even-parity magnets, e.g., $m^d_{\mathbf{k}}$ in Eq. (2), we arrive at $\mathcal{U}^\dagger[\Gamma(0)]^{-1}\mathcal{U} = \sum_{\mathbf{k}}(g^R_{\mathbf{k}}g^A_{\mathbf{k}})^{-1}M(\mathbf{k})$, where because $(g^R_{\mathbf{k}}g^A_{\mathbf{k}})^{-1}$ is diagonal, the Cooperon gaps is determined by the matrix $M(\mathbf{k})$, which reads

$$\begin{bmatrix} \lambda^2k^2 & 0 & -\lambda^2k_-^2 & 0 \\ 0 & 2m^2_{\mathbf{k}} + 2\lambda^2k^2 & 0 & 0 \\ -\lambda^2k_+^2 & 0 & \lambda^2k^2 & 0 \\ 0 & 0 & 0 & 2m^2_{\mathbf{k}} \end{bmatrix}. \tag{8}$$

When spin-orbit coupling λ is zero, $M(\mathbf{k})$ becomes diagonal and the λ^2k^2 terms vanish, leading to closing of the Cooperon gaps of two triplet channels $|1,1\rangle$ and $|1,-1\rangle$. Hence, WL dominates in even-parity magnets when there is no spin-orbit coupling. When spin-orbit coupling is turned on, it can be shown that the matrix $\mathcal{U}^\dagger[\Gamma(0)]^{-1}\mathcal{U}$ has only positive eigen values, which means that all four channels have finite Cooperon gaps so there is no WL or WAL.

For odd-parity magnets, e.g., $m^p_{\mathbf{k}}$ in Eq. (2), the matrix $M(\mathbf{k})$ that determines the Cooperon gaps reads

$$\begin{bmatrix} 2m^2_{\mathbf{k}} + \lambda^2k^2 & i\sqrt{2}\lambda m_{\mathbf{k}}k_- & -\lambda^2k_-^2 & 0 \\ -i\sqrt{2}\lambda m_{\mathbf{k}}k_+ & 2\lambda^2k^2 & -i\sqrt{2}\lambda m_{\mathbf{k}}k_- & 0 \\ -\lambda^2k_+^2 & i\sqrt{2}\lambda m_{\mathbf{k}}k_+ & 2m^2_{\mathbf{k}} + \lambda^2k^2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \tag{9}$$

where the 0 matrix elements mean that the Cooperon gap of the singlet channel $|0,0\rangle$ does not couple to the triplet channels and always has a vanishing Cooperon gap to contribute a WAL channel. At zero spin-orbit coupling λ, the gap of the triplet channel $|1,0\rangle$ also vanishes, giving rise to a WL channel that cancels the WAL of the singlet channel. Hence, there is no WL or WAL when there is no spin-orbit coupling [69]. At finite spin-orbit coupling λ, it can be proved that the upper 3×3 block of $\mathcal{U}^\dagger[\Gamma(0)]^{-1}\mathcal{U}$ has only positive eigen values, hence, all the triplet Cooperon gaps are finite, leading to the suppression of WL. As a result, WAL of the singlet channel dominates in the odd-parity magnets in the presence of spin-orbit coupling.

Magnetoconductivity and experimental implications - Below we show that WL of even-parity d-wave altermagnets and WAL of odd-parity p-wave magnets can be measured by the low-field magnetoconductivity.

For d-wave altermagnets characterized by $m^d_{\mathbf{k}}$ in Eq. (2), the quantum interference correction to conductivity can be calculated with the help of the diagrams in Fig. 2 (b) as

$$\delta\sigma^d_{xx} = -\frac{2e^2\ell^2}{h}\int\frac{d^2\mathbf{q}}{(2\pi)^2} \times \left[\frac{1}{\frac{4\ell^2}{3\ell_{so}^2} + \ell^2(q_x^2 + \epsilon^2q_y^2)} + \frac{\epsilon^2}{\frac{4\ell^2}{3\ell_{so}^2} + \ell^2(\epsilon^2q_x^2 + q_y^2)}\right], \tag{10}$$

where $\ell = \sqrt{D_d\tau}$, $\ell_{so} = \sqrt{D_d\tau_{so}}$, $\ell_\phi = \sqrt{D_d\tau_\phi}$ are the mean free path, spin-orbit relaxation length,

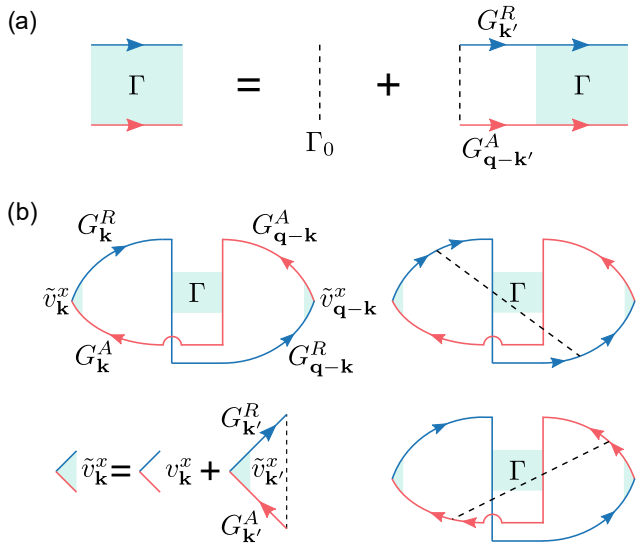


FIG. 2. The diagrams for the quantum interference correction to conductivity [68–74]. The arrowed solid lines and dashed lines represent the Green's functions $G^{R/A}$ and impurity scattering, respectively. (a) The Bethe-Salpeter equation for Cooperon Γ of maximally crossed diagrams. (b) The bare and two dressed maximally crossed diagrams give the quantum conductivity correction. $\tilde{v}$ is the vertex correction of the velocity v by ladder diagrams.

and phase coherence length, respectively, the diffusion constant $D_d = (v_F^d)^2\tau/2$, the Fermi velocity $v_F^d = \sqrt{4(a+J)E_F/\hbar^2}$, and $\epsilon^2 = (a+J)/(a-J)$, a and J depict the parabolic band dispersion defined in Eq. (1) and anisotropy induced by exchange interaction defined in Eq. (2), respectively. For convenience, we describe the spin-orbit coupling in terms of impurity scattering, i.e., spin-orbit scattering. The two terms in Eq. (10) stand for the two WL channels that can be gapped out by the spin-orbit scattering if the Cooperon gaps $4\ell^2/3\ell_{so}^2$ are finite, consistent with the above symmetry analysis and Cooperon gap calculations. By physically considering the lower and upper bounds as $1/\ell_\phi$ and $1/\ell$ in Eq. (10), we can find the quantum interference correction to conductivity at zero magnetic field as $\delta\sigma_{xx}^d(0) \simeq -\alpha e^2/(2\pi^2\hbar)\ln(1+\ell_2^2/\ell^2)$ and at finite z-direction magnetic field B as $\delta\sigma_{xx}^d(B) \simeq \alpha e^2/(2\pi^2\hbar)[\Psi(\ell_B^2/\epsilon\ell_2^2 + 1/2) - \ln(\ell_B^2/\epsilon\ell_1^2)]$, where $\alpha \equiv (1+\epsilon^2)/2\epsilon$, the magnetic length $\ell_B = \sqrt{\hbar/(4eB)}$, $1/\ell_1^2 \equiv 1/\ell^2 + 4/3\ell_{so}^2 + 1/\ell_\phi^2$, $1/\ell_2^2 \equiv 4/3\ell_{so}^2 + 1/\ell_\phi^2$. $\Psi(x)$ is the digamma function $\Psi(x) = -\gamma - 1/x - \sum_{k=1}^{\infty}[1/(x+k) - 1/k]$ with the Euler constant $\gamma \approx 0.57721$ [75, 76]. In the limit of vanishing spin-orbit scattering, i.e., $1/\ell_{so} \to 0$, Eq. (10) describes the WL at $B = 0$, which can be measured by the magnetoconductivity $\Delta\sigma_{xx}^d(B) \equiv \delta\sigma_{xx}^d(B) - \delta\sigma_{xx}^d(0)$ and

$$\Delta\sigma_{xx}^d(B) \simeq \alpha\frac{e^2}{2\pi^2\hbar}\Big[\Psi\Big(\frac{\ell_B^2}{\epsilon\ell_\phi^2} + \frac{1}{2}\Big) - \ln\frac{\ell_B^2}{\epsilon\ell_\phi^2}\Big], \qquad (11)$$

In contrast, if spin-orbit scattering is strong, i.e., $\ell_\phi, \ell_B \gg \ell_{so} \sim \ell$, the system turns to a diffusive metal at $B = 0$, then the magnetoconductivity nearly vanishes, i.e., $\Delta\sigma_{xx}^d(B) \to 0$, as shown in Fig. 3(a).

For p-wave magnets characterized by $m_{\mathbf{k}}^p$ in Eq. (2), the quantum interference correction to conductivity can be found as

$$\delta\sigma_{xx}^p = -\frac{2e^2\ell^2}{h}\int\frac{d^2\mathbf{q}}{(2\pi)^2}\Big(\frac{1}{\frac{4\ell^2}{3\ell_{so}^2} + \ell^2q^2} - \frac{1}{\ell^2q^2}\Big), \qquad (12)$$

where $\ell = \sqrt{D_p\tau}$, $\ell_{so} = \sqrt{D_p\tau_{so}}$, $\ell_\phi = \sqrt{D_p\tau_\phi}$ are the mean free path, spin-orbit relaxation length, and phase coherence length, respectively, the diffusion constant $D_p = (v_F^p)^2\tau/2$, the Fermi velocity $v_F^p = \sqrt{4aE_F/\hbar^2}$. The first and second terms in Eq. (12) describe a WL channel that can be gapped out by the spin-orbit scattering and a WAL channel, respectively. The quantum interference correction to conductivity at zero magnetic field is found as $\delta\sigma_{xx}^p(0) \simeq e^2/(4\pi^2\hbar)[\ln(\ell_\phi^2/\ell^2) - \ln(1+\ell_2^2/\ell^2)]$, where the first and second terms represent the WL and WAL contributions, respectively. The quantum interference corrections to conductivity at finite z-direction magnetic field is found as $\delta\sigma_{xx}^p(B) \simeq e^2/(4\pi^2\hbar)\{\Psi(\ell_B^2/\ell_2^2 + 1/2) - \ln(\ell_B^2/\ell_1^2) - [\Psi(\ell_B^2/\ell_\phi^2 + 1/2) - \ln(\ell_B^2/\ell^2)]\}$. In the case of strong spin-orbit scattering, i.e., $\ell_\phi, \ell_B \gg \ell_{so} \sim \ell$, the WL channel is gapped out, only the WAL channel survives, which can be measured by the magnetoconductivity $\Delta\sigma_{xx}^p(B) \equiv \delta\sigma_{xx}^p(B) - \delta\sigma_{xx}^p(0)$ and

$$\Delta\sigma_{xx}^p(B) \approx \alpha\frac{e^2}{2\pi^2\hbar}\Big[\Psi\Big(\frac{\ell_B^2}{\ell_\phi^2} + \frac{1}{2}\Big) - \ln\frac{\ell_B^2}{\ell_\phi^2}\Big], \qquad (13)$$

with $\alpha = -1/2$. By comparison, in the limit of vanishing spin-orbit scattering, i.e., $1/\ell_{so} \to 0$, the magnetoconductivity vanishes, i.e., $\Delta\sigma_{xx}^p(B) \to 0$, as shown in Fig. 3(a).

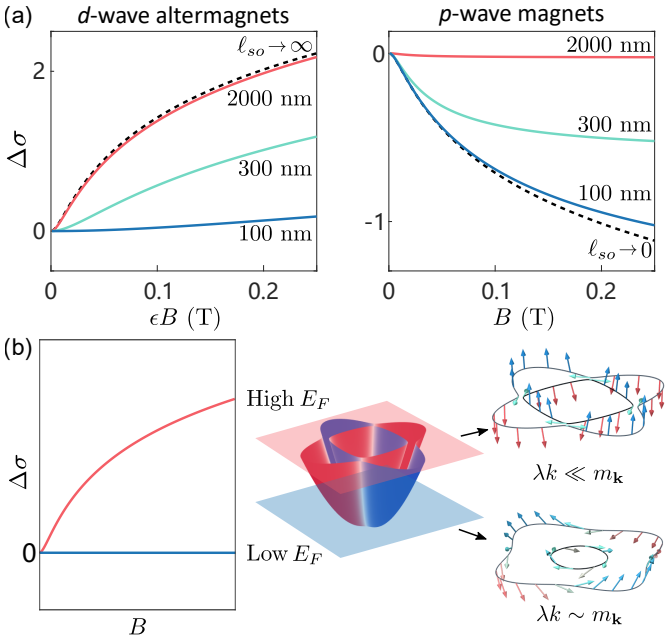


FIG. 3. (a) Magnetoconductivity $\Delta\sigma$ of the d-wave altermagnets [in units of $(1+\epsilon^2)e^2/(4\epsilon\pi^2\hbar)$] and p-wave magnets [in units of $e^2/(2\pi^2\hbar)$] at different strengths of spin-orbit scattering. A shorter spin-orbit relaxation length ℓ_{so} means stronger spin-orbit scattering. Spin-orbit scattering suppresses WL in the d-wave altermagnets and activates WAL in the p-wave case. Parameters are chosen as $\ell = 10$ nm and $\ell_\phi = 400$ nm. (b) For d-wave altermagnets, the Fermi surface is dominated by $m_{\mathbf{k}}$ for high Fermi energy, corresponding to WL. For low Fermi energy, $m_{\mathbf{k}}$ become relatively close to the spin-orbit coupling. WL is suppressed and the system is in a diffusive metal state.

Discussions - Besides directly tuning the spin-orbit coupling strength λ, the relative magnitude of $m_{\mathbf{k}}$ and λk can also be controlled by shifting the Fermi surface. For a higher-order magnetic splitting $m_{\mathbf{k}} \propto k^n$ $(n > 1)$, lowering the Fermi energy toward the band bottom reduces $m_{\mathbf{k}}$ faster than λk. As illustrated in Fig. 3(b), the system can therefore evolve from a regime with $\lambda k \ll m_{\mathbf{k}}$, where WL is pronounced, to one with $\lambda k \sim m_{\mathbf{k}}$, where the WL Cooperon gaps are substantially enhanced and the system approaches a diffusive-metal regime. Thus, tuning Fermi energy provides an alternative route to control quantum interference.

For p-wave magnets, two triplet Cooperon channels $|1,1\rangle$ and $|1,-1\rangle$ may also give rise to WL at finite

wavevectors $\pm\mathbf{q}_0 = \pm J/(2a)\hat{x}$, because the spin-up and spin-down Fermi surfaces of p-wave magnets are shifted in opposite directions. In this case, p-wave magnets may have totally 3 triplet Cooperon channels and 1 singlet Cooperon channel in the absence of spin-orbit coupling to exhibit WL. Spin-orbit coupling can suppress the 3 WL channels, leading to a crossover to WAL. Nevertheless, in realistic lattices, the warping of the Fermi surface may suppress the $|1,1\rangle$ and $|1,-1\rangle$ WL channels by breaking the $\mathbf{k} \to \pm\mathbf{q}_0 - \mathbf{k}$ symmetry.

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